

consider the impact of the packaging architecture and materials on the optical and electrical performance [2]. Different types of co-packaged architectures are being explored and researched [8-12] as seen in Figures 2, 3 and 4. Recently, the research focus is moving towards splitting the photonics die into an optical and electrical component as two separate dies stacked in 3D to drive shorter interconnections. One of the key requirements for any integrated co-packaged architecture is the photonics die to have an overhang to be able to attach the optical connector. Warpage and flatness of the Fiber Assembly Units (FAUs) is one of the key concerns for the fiber-based optical interconnects to attach to the photonics die as it directly impacts the insertion loss and bandwidth. With thin die requirements and complex architectures, it is critical to optimize the assembly conditions to mitigate the warpage [13, 14]. Meeting the quality and reliability needs for the semiconductor industry is another key concern for CPO architectures [15].

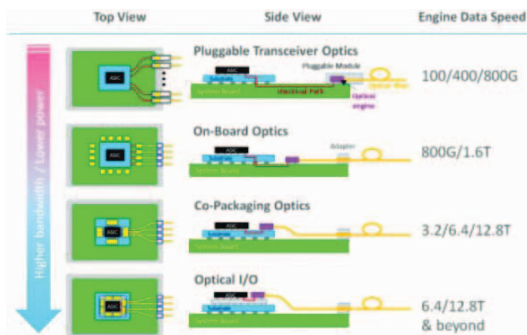


Fig. 2. Evolution of fiber optics [11]

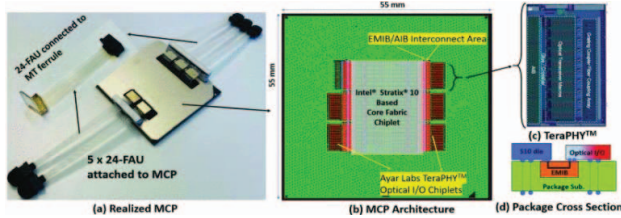


Fig. 3. Example of CPO (a) Fully assembled optical MCP (b) MCP architecture (c) the optical chiplet layout view (d) cross section of the MCP [12]

To execute the demonstration of an optical co-package prototype, DARPA awarded Intel with the Photonics in the Package for Extreme Scalability (PIPES) Program and Common Heterogeneous Integration and IP Reuse Strategies (CHIPS) programs [16-19]. The prototype developed under this initiative was a multi-chip package (MCP) featuring an innovative open cavity bridge substrate that incorporates Intel's cutting-edge Embedded Silicon Bridge (EMIB) technology. In this paper, we will review the assembly of a novel open cavity bridge substrate with EMIB architecture connecting the field programmable gate array (FPGA) die to data converter dies on one side and 3D EIC-PIC stacks on the other side. The optical interconnect is achieved through fiber-array unit connected to the edge of the photonics die with butt-coupling technology.

The fundamental understanding of materials, process and tool interaction is required to define the assembly process for this complex heterogeneous architecture [20]. This paper will overview the thermomechanical modeling, design and architecture details, and highlight the key benefits of open cavity bridge architecture. The paper will then discuss the major processes and their associated challenges such as (a) photonics die (PIC) thinning and singulation while maintaining integrity of the optical edge, (b) EIC-PIC stack assembly with PIC die overhang, and (c) package assembly constraints considering the thermal and mechanical stability with an open cavity bridge substrate. Through extensive process development, failure analysis and characterization, CHIPS prototype MCP was successfully assembled with the first-of-its-kind open cavity bridge architecture to enable excellent optical performance.

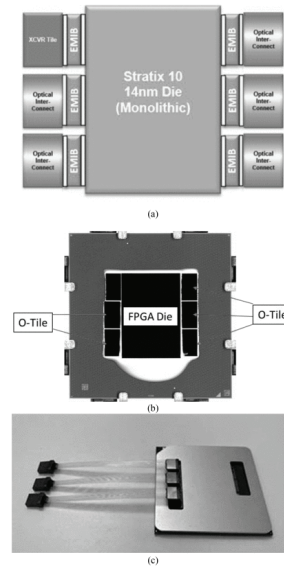


Fig. 4. Example of CPO (a) Schematic of an MCP with 5 optical O-Tiles (b) MCP Package with 5 O-Tiles (c) Integrated Heat Spreader (IHS) design that allows for top-side fiber attach of a FAU [2]

II. PACKAGE ARCHITECTURE DETAILS

The CHIPS program aims to redefine IP reuse methodologies to address the issues of design expenses associated with sophisticated SoCs, especially for niche applications and enhance system adaptability while minimizing design efforts for future innovations. Using the concept of IP reuse, this prototype leverages a 65 x 65 mm large form factor package with an FPGA or a general-purpose unit (XPU) with EMIB bridges on both sides connecting to two data converter dies on one side and three EIC-PIC die stacks on the other side. The PIC die is embedded in the cavity of the substrate with the PIC edge connecting to FAUs via butt-coupling interconnect technology. This MCP is a collaboration between Cornell University for EIC design, Columbia and AIM photonics for

PIC design, State University of New York for PIC manufacturing and Tyndall Institute for fiber attach (Fig. 5). Some of the key motivations for this architecture is the direct EIC-PIC interconnections with lesser electrical transitions, easier access of the PIC edge for fiber attach, no mechanical overhang compared to groove-type fiber interconnects and better strain relief. Additionally, the open cavity architecture allows for traditional simplified processing of the EIC die as opposed to 3D EIC-PIC without cavity where the EIC die acts as a silicon interposer and must be thinned to allow for TSV reveal and bumping on the silicon for package side interconnects.

Despite the advantages described above, this architecture comes with its own set of challenges. Some of the key risks during assembly include – delamination due to substrate cavity manufacturing, high warpage on the PIC die posing challenges during fiber butt-coupling, process and handling risks in attaching the EIC-PIC 3D stack with an overhang on the EIC die, PIC thinning and singulation while maintaining the edge integrity. To address these risks, modeling is key to understand the failure modes and plan for the MCP assembly [21].

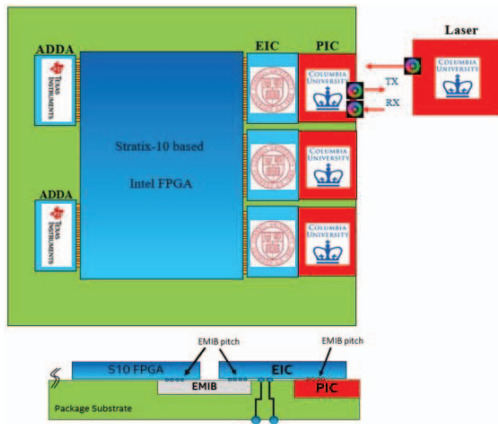


Fig. 5. Schematic of the CHIPS multi-chip package architecture

III. III. MODELING

A parametric model was developed to understand the key failure modes associated with the identified risks and assist with defining the assembly process and materials. Package warpage at room temperature and 80°C was evaluated with convex warpage prediction (Fig. 6). Additionally, the PIC die warpage and tilt was evaluated for front edge to assess the high-risk area for fiber attach. The center PIC die is shown to be higher risk for warpage and tilt for the front edge warpage that attaches to the FAU (Fig. 7).

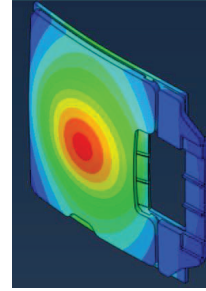


Fig. 6. Model-predicted package warpage at room temperature.

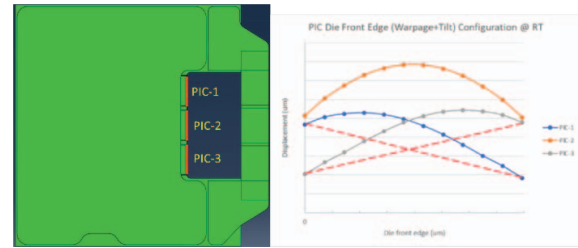


Fig. 7. Model-predicted PIC front edge warpage and tilt. RT stands for Room Temperature.

The substrate was also modeled for assessing the delamination risk. Overall, the substrate is expected to be at high thermal stress risk with the local gradient in the EMIB regions being the worse (Fig. 8).

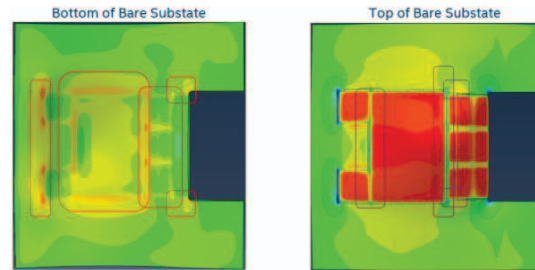


Fig. 8. Model-predicted bare substrate delamination risk at 260C temperatures. The red boxes indicate high stress areas

IV. IV. ASSEMBLY PROCESS AND CHALLENGES

A. Photonics Die Preparation

One of the first challenges of this assembly was developing a novel die preparation process to enable frontside plasma trenching on PIC die. Frontside plasma trenching [22, 23], fabricated by SUNY Polytechnic Institute is critical in achieving die edge quality for successful fiber attach and performance. The process etches the full street width around each die at a precise, adjustable depth with a minor wall chamfer and rounding of trench base. To enable thinning, singulation and die pick processes from original equipment and collaterals a first-of-a-kind temporary carrier was developed. Carrier assured protection of active die side, no die movement

or lost during processing and meeting ESD requirements. Overall flatness exceeded wafer tolerances to achieve uniform die thickness of $\pm 2 \mu\text{m}$. Carrier material and release method greatly reduce stress and residue, enabling ultra-thin die (50 μm) processing and removed die cleaning operations before bonding. Plasma trenched wafer is bonded to carrier with trench face down. Bonded wafer is thinned beyond plasma trench depth which results in die singulation. Thinning operation may continue at die level until desired thickness is reached (Fig. 9). Die on carrier completes release process and are transferred from equipment on carrier. Dies are picked from carrier and can be stored on carrier for extended time with no limits to number of pick cycles.

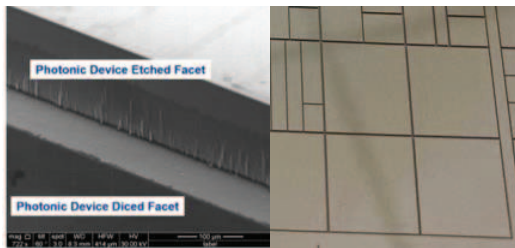


Fig. 9. Plasma Trenching (Left - Ack: SUNY Poly) and Post thinning and singulation (Right)

B. Optical 3D Stack Assembly

The 3D EIC-PIC die stacks were assembled prior to assembly on to the substrate instead of attaching the PIC die in the cavity first, and then bonding the EIC die simultaneously to the PIC and substrate. Stack assembly on substrate allows for tolerances incoming from die height, PIC and substrate bump variations. For the 3D die stack assembly, the key challenges are discrete singulated die to die bonding, offset die stack attach and precision underfill for stack and open cavity. To enable die to die bonding a novel bonding method and carrier were developed. As with die preparation, this second temporary carrier enabled die to die bonding on original equipment and collaterals. Carrier was capable of thermal and bonding forces for die bonding, assured no die movement and met ESD and flatness requirements. Carrier could be modified to different form factors and allow multiple dies reconstitution on single carrier with user defined die position and pitch. Die to die bonding process is a “Hot Tack” thermal compression bonding (TCB) with formic acid reflow (FAR). This process allows a temporary weak bond through minimum wetting and mechanical deformation of solder bumps. After TCB die stacks go through FAR which cleans surfaces, allows self-alignment, and completes full solder collapse and wetting. In addition, FAR process improved defects of electroplating and solder bumping process of individual PIC die. Inline TCB FAR bonding process was used for all die bonding to substrate. Bonding placement and tilt of offset PIC/EIC die stack within tight open cavity tolerances was controlled while minimizing thermal exposure to substrate (Fig. 10). The final challenge was underfill (UF) delivery with several restricted areas, multiple

surfaces and features with limited access. Underfill was done with a jet valve requiring several dispense locations and passes to precisely control flow and coverage. Underfill material was filled under EIC die attached to substrate and between EIC and PIC die without flow into restricted area on PIC die. In the cavity, material was filled under PIC die without bridging between PIC dies, without material rollup on restricted area or exceeding cavity edge (Fig. 11).

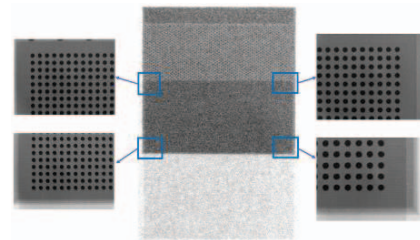


Fig. 10. 2D-Xray of die-to-die bonding

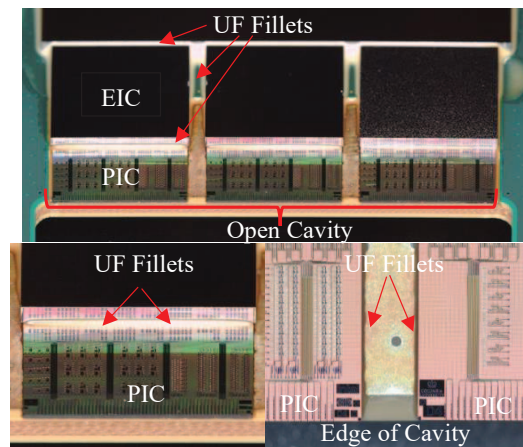


Fig. 11. Underfill of EIC/PIC die stacks in Open Cavity

C. Package Assembly

The package assembly process was done using a standard flow: TCB $\rightarrow$ UF dispense $\rightarrow$ IHS lid-attach $\rightarrow$ ball-grid array (BGA) attach as shown in Fig. 12. One of the key challenges observed during package assembly is the thermal sensitivity of the substrate. With an open cavity, cutout and a large form factor, any high temperature thermal exposure step resulted in delamination (Fig. 13). This could be attributed to the in-plane surface tension stress gradient experienced in the substrate as shown in the substrate stress modeling.

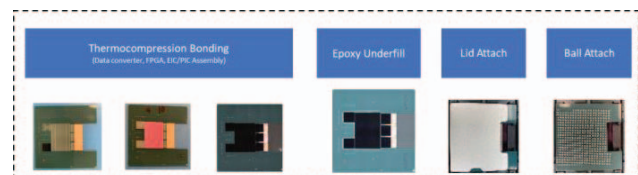


Fig. 12. Package assembly process flow.

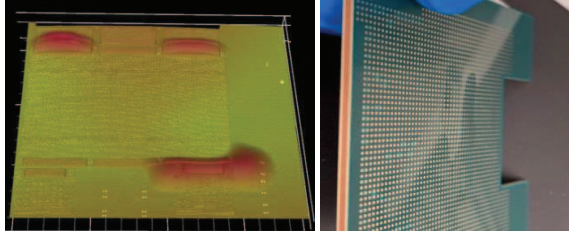


Fig. 13. Substrate delamination post thermal exposure.

To understand the quality of the incoming substrate, a thermal bake step at solder melting temperatures was implemented to pre-screen the substrates prone to delamination. Yield loss due to delamination was observed in ~50% of the substrates. With each subsequent thermal exposure step during chip-attach through thermal compression bonding, integrated lid-attach and ball attach, the yield loss increased to ~90%. The major contribution to the yield loss was observed post the pre-screening and TCB process, indicating high sensitivity of the substrate to thermal exposure. The next step was to eliminate high thermal exposure from the above processes.

Optimization of the thermal profile during TCB [24], and all subsequent thermal processes is critical to mitigate delamination. With a lower thermal exposure process, no delamination was achieved. To verify the assembly process robustness, extensive characterization was performed in the form of 3D X-Ray to check for bump alignment, c-Mode Scanning Acoustic Microscopy (C-SAM/TSAM) and optical imaging to check for voiding and delamination, validated through cross-sections (Fig. 14 and Fig. 15). After successful chip-attach assembly process without delamination, the lid and ball attach process were crucial to monitor. No delamination was observed in these processes, indicating the maximum contribution and initiation of delamination failure was from the high thermal exposure during the previously used TCB process, worsening in subsequent thermal steps. With elimination of delamination through the developed TCB process, the subsequent process steps did not further initiate delamination. This enabled a robust assembly process without the need for substrate redesign.

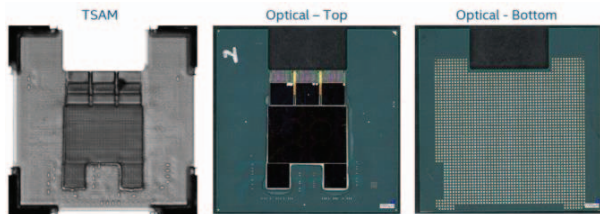


Fig. 14. Delamination and voiding characterization of the package assembly process through TSAM and optical inspection.

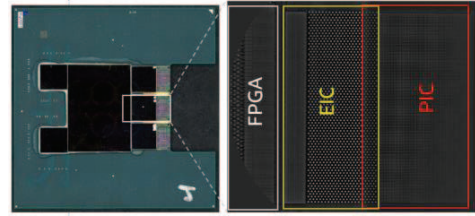


Fig. 15. 3D X-Ray inspection depicting good bump alignment between FPGA-EIC-PIC dies during the assembly process.

For the lid attach process, additional considerations beyond delamination were identified. Selection of the right thermal interface material (TIM) was critical – solder vs liquid metal vs polymer TIM. Polymer TIM was chosen to ensure no electrical interference to the EIC-PIC stack, especially due to high likelihood of the TIM material flowing in between the EIC and PIC dies. Additionally, contamination to the PIC die edge was a high-risk factor to optical fiber attach assembly. Through material selection and process optimization, the above risks were mitigated. The subsequent ball-attach step was also processed with no solder voids and excellent alignment. Low-temperature solder was used as the ball-grid array bumps, due to high thermal sensitivity of the materials and components in the photonics packages, along with other advantages of lower package warpage and stress, etc. [25, 26]. The MCPs were then sent to Tyndall Institute for fiber attach assembly. Tyndall was able to assemble the FAUs through butt-coupling interconnection with good mechanical integrity of the FAUs as shown in the Fig. 16 below.

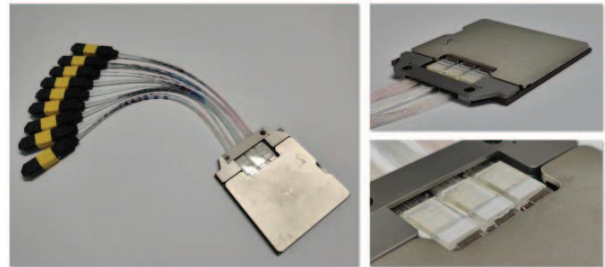


Fig. 16. Demonstration of final MCP package post fiber-attach process (Ack: Tyndall Institute).

V. CONCLUSION

Through modeling, process optimization and extensive failure analysis, the CHIPS MCP assembly was demonstrated with good bump alignment, no voiding or delamination and minimum warpage for successful fiber attach assembly at Tyndall.

Co-packaged optics is the future of microelectronics packaging to achieve high bandwidths with increased power efficiency. The proof-of concept (POC) MCP demonstrated in this paper in addition to POCs being highly researched in this area show the promising nature of integrating photonics within a package. Combining such with advanced packaging architectures such as Foveros and Foveros Direct can enable more power efficient and higher bandwidth photonic packages. Key challenges for the widespread adoption of CPO are the throughput, yield, affordability and cost associated with optical connectors, package

assembly and test [27, 28]. Board-level and system-level reliability of CPO architectures is another key area with limited data [29, 30]. Future CPO research would need to address and optimize system-level reliability and develop architectures that can maximize yield and throughput [2, 31, 32].

V. ACKNOWLEDGMENT

The authors would like to thank Xiaoqian Li, Omkar Karhade, Chia-Pin Chiu for the early development of this multi-chip package prototype architecture and Yuxin Fang for the substrate evaluation. The authors would also like to thank Cornell University for EIC design, Columbia and AIM photonics for PIC design, State University of New York for PIC manufacturing and bumping. For the photonics fiber array unit assembly, we would like to thank Tyndall Institute for extensive co-development of strain-relief mechanisms and FAU butt-coupling process with successful demonstration. The research on integrating the CHIPS MCP was developed with funding from the Defense Advanced Research Projects Agency (DARPA) as part of the PIPES program. The views, opinions and/or findings expressed are those of the author and should not be interpreted as representing the official views or policies of the Department of Defense or the U.S. Government.

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